TalentMatch ATS — System Guide

Resume Folder Access, File Processing & Local AI Information Extraction

ARCHITECTURE & USER GUIDE

Core Flow: When a user connects a local resume folder, TalentMatch operates 100% locally on their computer. It extracts text from all resumes (PDF, Word, Text), runs local AI parsing (via Ollama or built-in engine), compiles complete candidate dossiers, and scores job matches without uploading sensitive files to any cloud.

1. What Happens When Folder Access Is Granted

1 Native OS Folder Picker

User clicks **"Connect Folder"**. The operating system prompts for folder selection with read-only access. Files remain entirely on the local device.

2 Recursive Scan & File Discovery

The scanner looks through all root files and subdirectories. Supported formats (`.pdf` , `.docx` , `.doc` , `.txt`) are queued. Other formats are safely ignored.

3 In-Memory Text Extraction

- **PDF:** Reconstructed spatially from top-to-bottom and left-to-right (avoids broken columns).
- **Word (.docx):** Extracted via Mammoth directly into clean structured text.

4 AI Clinical Extraction (Ollama / Local LLM)

Resume text is processed by a local model (such as `llama3.2:3b` at `localhost:11434`) into a structured JSON record.

5 Profile Storage & Job Match Calculation

Profiles are saved into local storage and instantly scored against open requisitions to show fit percentage, missing qualifications, and experience match.

2. Exactly What Information Is Extracted

Extracted Field	Description & Clinical Logic	Sample Output
Candidate Name	Full Name parsed from header or contact block	`Sarah Jenkins, RN`
Contact Info	Direct phone number and verified email address	`sarah.j@email.com · (555) 234-5678`
Position & Specialty	Primary title & unit specialty (ICU, ER, OR, etc.)	`ICU Registered Nurse` (Specialty: `Critical Care`)
Experience Tenure	Total calculated professional clinical tenure (years)	`5 Years`
Current Location	City and State of primary residence	`Austin, TX`
Travel / Work History	Prior states or metro areas where candidate practiced	`["Denver, CO", "Phoenix, AZ", "Seattle, WA"]`

Extracted Field	Description & Clinical Logic	Sample Output
Licenses & Certs	Clinical certifications & active state licenses	`["BLS", "ACLS", "CCRN", "Compact State RN"]`
Professional Summary	2–3 sentence synthesized background profile	`"Experienced ICU RN with 5+ years handling high-acuity trauma and telemetry..."`
Career Timeline	Structured history: Facility, Role, Dates, Scope	`St. David's Medical (2021–Present, Staff RN)`

💻 Windows Hardware Specs

- **Recommended Local LLM:** `llama3.2:3b` (Ollama)
- **RAM:** 8 GB Minimum (16 GB for faster batch)
- **CPU:** Intel Core i5 / AMD Ryzen 5 or higher
- **Speed:** 10–25s per resume (CPU) / 2–4s (NVIDIA GPU)
- **Cost & Privacy:** 100% Free, zero cloud data transfer

⚠️ Key Extraction Caveats

- **Scanned Image PDFs:** Image-only photocopies without text layers cannot be read (requires selectable PDF).
- **Legacy .doc format:** Older binary formats may lose formatting; `.docx` and `.pdf` are best.
- **Browser CORS:** For web mode, run `set OLLAMA_ORIGINS=*` before starting Ollama, or use the Desktop .exe app.